KALAHARI LOGISTICS LTD

Wireless Networking Solutions — Assignment 1

INDIVIDUAL REFLECTION

Member: Member 1 — Namibia & Network Lead

Role: Namibia research, WAN/wireless analysis and network-design coordination

This reflection is based on the requirements and work of the group project and is suitable for the individual reflection section.

As the Namibia and Network Lead, my main responsibility in this assignment was researching Namibia's wireless and WAN infrastructure and contributing to the overall network design for Kalahari Logistics Ltd. The project gave me a better understanding of how networking technologies are applied in a real business environment. I learned that Namibia's large geographical area, low population density and distance between towns can make terrestrial network deployment more challenging and expensive. This helped me understand why fibre, cellular networks, leased lines, VPNs and satellite connectivity may be selected according to location and business requirements.

One of the most challenging parts was deciding how to justify the WAN technologies in our proposed design. I learned that network design is not simply about selecting the fastest technology. Distance, availability, reliability, cost, infrastructure and future scalability must all be considered. This was particularly important when considering connectivity to remote areas and the global site.

The SADC perspective also taught me that cross-border networking introduces additional challenges. Differences in regulations, infrastructure availability, roaming costs and infrastructure sharing can affect multinational organisations. I realised that networking is influenced not only by technical requirements but also by economic, regulatory and regional factors.

Working with Cisco Packet Tracer helped me connect the research to practical networking. I developed a better understanding of how routers, switches, access points, IP addressing and WAN links work together to create an interconnected network. Testing the topology also showed me the importance of systematic troubleshooting because a small error in an IP address, interface or routing configuration can prevent communication between sites.

Another important lesson was teamwork. Because each group member had a specific responsibility, I needed to communicate my research and network requirements clearly, especially when coordinating the overall design. I also gained a better appreciation for GitHub and version control as tools for collaborative IT work. The project showed me that successful network engineering requires both technical skills and effective communication.

If Kalahari Logistics doubled its number of branches, I would improve the design by implementing more structured subnetting, VLANs, stronger security controls, redundant WAN connections and a scalable dynamic-routing or SD-WAN approach. I would also consider centralised monitoring and management so that additional branches could be integrated without making the network unnecessarily difficult to manage.

Overall, the most valuable lesson I gained from this assignment is that network engineering involves much more than configuring devices. A network engineer must understand the organisation's

requirements, geographical environment, available technologies, security needs, cost and scalability before selecting a solution. This project strengthened my understanding of Namibia’s networking environment and showed me how wireless networking, WANs, IP addressing, routing and network design can be applied to a realistic business scenario. It has therefore helped bridge the gap between the concepts I learn in class and the skills I would need in a real networking environment.